

Lecture quiz 1 grades released later this week on Gradescope

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5: Which of these equations represents surface

A represents a sphere, but the 1st graph is showing that the surface is unbounded, so it's not A.

The 2nd graph is showing the cross section given by intersecting the  $xz$ -plane, which is where  $y=0$ .

So when  $y=0$ , the surface is an upwards parabola, something like  $z=cx^2$ ,  $c>0$ .

Now let's see what graph the other 3 choices give when  $y=0$ :

B:  $z=1-x^2-2y^2=1-x^2$ , downward parabola

C:  $z^2 = x^2 + y^2 = x^2 \Rightarrow z = \pm x$ , 2 lines

D:  $z = 2x^2 - y^2 = 2x^2$ , upwards parabola

So D is the only one also showing an upwards parabola when  $y=0 \Rightarrow \boxed{\text{ans} = D}$

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7: Find  $\lim_{t \rightarrow 1} r(t)$  where

$$r(t) = \left\langle \frac{t^2 - 1}{t^2 - 3t + 2}, \frac{t - 1}{\sqrt{t+3} - 2}, \frac{\sin(t-1)}{t-1} \right\rangle$$

Recall: limit of vector function exists only if every component limit exists

1st:  $\lim_{t \rightarrow 1} \frac{t^2 - 1}{t^2 - 3t + 2} = \lim_{t \rightarrow 1} \frac{(t-1)(t+1)}{(t-1)(t-2)}$

$$= \lim_{t \rightarrow 1} \frac{t+1}{t-2} = \frac{1+1}{1-2} = \underline{-2}, \text{ noting}$$

that we can cancel  $(t-1)$  since the behavior at  $t=1$  doesn't matter for the limit, so  $\frac{t-1}{t-1} = 1$  whenever  $t \neq 1$ .

2nd: 
$$\frac{t-1}{\sqrt{t+3}-2} = \frac{(t-1)(\sqrt{t+3}+2)}{\sqrt{t+3}^2 - 2^2} =$$
  

$$\frac{(t-1)(\sqrt{t+3}+2)}{t-1} = \sqrt{t+3}+2$$

whenever  $t \neq 1$ . So limit is

$$\lim_{t \rightarrow 1} (\sqrt{t+3}+2) = \sqrt{1+3}+2 = \underline{4}$$

3rd: 
$$\lim_{t \rightarrow 1} \frac{\sin(t-1)}{t-1} = \lim_{t \rightarrow 0} \frac{\sin t}{t}$$

by using substitution  $t \rightarrow t+1$ ,

and this limit is 1 from calc I  
or any other prior source.

So the 3 limits are  $\frac{2}{3}, 4, 1$ , so

$$\lim_{t \rightarrow 1} r(t) = \boxed{\langle -2, 4, 1 \rangle}.$$

6: Domain of  $\langle \ln(t+1), \frac{t}{4-t^2}, \sqrt{4-t} \rangle$

Recall: vector is defined if and only if

every component is defined.

1st:  $\ln(\dots)$  is defined when argument is positive, so  $t+1 > 0 \Rightarrow t > -1$

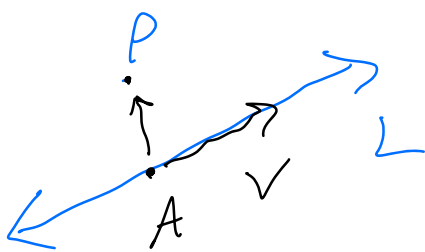
2nd:  $t$  &  $4-t^2$  always defined, so only issue with  $\frac{t}{4-t^2}$  is division by 0. So it's defined if  $4-t^2 \neq 0 \Rightarrow t \neq \pm 2$

3rd:  $\sqrt{\dots}$  is defined when argument is  $\geq 0$ , so  $4-t \geq 0 \Rightarrow t \leq 4$ .

Combine 1st & 3rd  $\Rightarrow -1 < t \leq 4$ . Use the 2nd, then  $t \neq 2$ . So the answer is

$$(-1, 2) \cup (2, 4]$$

3: Plane through  $P = (10, -1, 5)$  containing the line  $\frac{x}{4} = y + 6 = \frac{z}{5}$ .



This is an unused collection of information, so to solve this problem,

we need to translate to a common scenario.

Pick some point  $A$  on the line  $L$  and let  $\vec{v}$  be the direction of  $L$ . Then  $A, P$  are on the plane, and  $\vec{v}, \vec{AP}$  are vectors on the plane.

We want to find some normal vector  $\vec{n}$ . Since  $\vec{v}$  &  $\vec{AP}$  are 2 independent vectors in the plane, we can take  $\vec{n} = \vec{v} \times \vec{AP}$ .

In conclusion, need to find  $A, \vec{v}, \vec{v} \times \vec{AP}$ .

$y+6$  is part of the equations for  $L$  & it's convenient to solve equations when a term is 0, so let  $y = -6$ .

Then  $\frac{x}{4} = 0 = \frac{z}{5} \Rightarrow x = z = 0,$

so  $A = (0, -6, 0) \in L$ .

We can rewrite the line as

$$\frac{x}{4} = \frac{y+6}{1} = \frac{z}{5}, \text{ so}$$

$\vec{v} = \langle 4, 1, 5 \rangle$  is a choice for the

direction of the line from last week's content.

$$\overrightarrow{AP} = P - A = (10, -1, 5) - (0, -6, 0) = \langle 10, 5, 5 \rangle.$$

$$\vec{n} = \vec{v} \times \overrightarrow{AP} = \langle 4, 1, 5 \rangle \times \langle 10, 5, 5 \rangle$$

$$= 5(\langle 4, 1, 5 \rangle \times \langle 2, 1, 1 \rangle)$$

$$= 5\langle -4, 6, 2 \rangle$$

$$= 10\langle -2, 3, 1 \rangle, \text{ so}$$

$$\begin{array}{rcl} 4 & 1 & 5 \\ 2 & 1 & 1 \end{array} \begin{array}{l} \therefore -4 \\ \therefore 6 \\ \therefore 2 \end{array}$$

$\langle -2, 3, 1 \rangle$  is also a normal  $\Rightarrow$

$$-2(x-0) + 3(y+6) + 1(z-0) = 0$$

is an equation of the plane.

1: Parametric equations for intersection of  $2x + 3y + 5z = 7$ ,  $x - y + 2z = 3$ .

Recall: parametric equations have the form  $x = \dots$ ,  $y = \dots$ ,  $z = \dots$  with 1 variable.

We have 3 variables & 2 equations.

But every single time you use an equation, you can eliminate 1 variable.

Let's begin with 2nd equation since the coefficients are smaller:

$$3 = x - y + 2z \Rightarrow x = 3 + y - 2z$$

Let's substitute this into 1st equation:

$$7 = 2x + 3y + 5z = 6 + 2y - 4z + 3y + 5z$$

$$1 = 5y + z \Rightarrow z = 1 - 5y.$$

$$\text{Also note } x = 3 + y - 2(1 - 5y) = 1 + 11y.$$

Since everything is in terms of  $y$ , we can just let  $y$  be the parameter:

$$x = 1 + 11y, \quad y = y, \quad z = 1 - 5y, \quad y \in \mathbb{R}$$

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4: Does  $r(t) = \langle -2t, 2+7t, -1-4t \rangle$  intersect the plane  $4x + 9y - 2z + 8 = 0$

Recall: 2 objects intersect iff the equations governing them have simultaneous solutions

So let's take the equation for  $L$ , plug it into the plane equation, and see if

there are any solutions:

$$\begin{aligned} 0 &= 4(-2t) + 9(2+7t) - 2(-1-4t) + 8 = \\ &= -8t + 18 + 63t + 2 + 8t + 8 = 63t + 28 \\ &\Rightarrow t = -\frac{28}{63} = -\frac{4}{9} \end{aligned}$$

So there is an intersection point, which you can find by plugging  $t = -\frac{4}{9}$  back into  $r(t)$ .

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